

Large Language Models Can Be Easily Distracted by Irrelevant Context

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Abstract

Large language models have been greatly transformed in the field of natural language processing. For example, famous models such as GPT-4, DeepSeek, and Gemini have been changed a lot due to multiple prompting techniques, and they resemble human communication across various topics. However, a significant challenge they face is they get distracted with irrelevant content and this is the major problem nowadays with large language and we want to resolve this problem by applying some of the solutions which we can use by applying multiple prompting techniques and we can provide techniques like self-consistency, so we can improve the working of this large language model and they can handle irrelevant data nicely and give us more accuracy while handling irrelevant data.

1 Introduction

Large Language Models or LLMs are AI systems trained on massive datasets to generate understand human-like text. These large language models have the capabilities to demonstrate abilities in specific tasks like summarization, logic, reasoning, and problem solving. However, most of the time their performance is measured in a controlled setting or controlled environment where the provided information is relevant. So, the major problem is if the model sees any unnecessary detail, will it ignore them or will focus on solving the problem, or will it get confused. The issue is that the large-language model gets easily distracted with irrelevant content, noise in data, and distractors. And in real-world applications, the data is noisy. But for a perfect world for a large-language model, that is not the case. For a large-language model, the data should be relevant and in synchronized manner. But in real-world, it doesn't work like that. The data is messy. Data contains many distractors. So our large-language model should be robust enough to

handle this distraction effectively. In this specific decade, our reliability over large-language model is increasing drastically day by day. And if the LLM is not performing well in this specific task of handling irrelevant data, we can get errors in critical domains such as healthcare, where AI has to summarize medical records and do image mapping. So, if AI ignores the major symptoms of patient and test results and focuses on unrelated details in the report, which are not related to our major outcome, then it should be an issue and our reliability will decrease day by day. So without proper filtering mechanism, AI model risks making incorrect predictions or generating wrong responses, which can have serious consequences on various fields like finance and business, legal analysis, healthcare. And to resolve this particular problem, which is our main objective of this research paper, there are multiple ways we can test the capacity of large language model to ignore the irrelevant data and give us the highest state of accuracy. We can try multiple prompting techniques like chain of thought technique, zero-shot chain of thought, direct answer baseline, self-consistency. These are the various techniques which we can use to reduce the distractibility of the model. And we will see in the experiment which is the most effective technique you can use to make the model adapted to irrelevant content and which gives you higher accuracy than others.

2 Dataset

To test the model accuracy with irrelevant content, we decided to choose a dataset on Arithmetic Reason Gold standards and one of the most famous dataset which is used as a gold standard for Arithmetic Reasoning and logical problem, which is GSM-8K, which is a Grade-School Math dataset, which requires step-by-step reasoning to each correct answer. The dataset makes sure that model must perform logical and arithmetic reasoning in-

stead of relying moreover on pattern recognition or getting adapted to similar problem. But there is a major limitation in GSM-8K dataset, which is it only contains relevant information. It only contains clean problem which requires step-by-step problem solving technique. But in real world, problem often come with unnecessary details that can distract our large language model drastically and can have a big impact on the result. So, this brings us to the modification of GSM-8K version, which we call as GSM-IC, which stands for Grade School Math with Irrelevant Content. Since GSM-8K do not have any irrelevant content, we modified this version and designed to evaluate how LLMs handle irrelevant information. GSM-IC is a dataset deprived from GSM-8K, but with one of the major modification is that each problem contain one additional irrelevant sentence that does not contribute to the answer of the question or helps in solving the question.

3 Related Work

3.1 Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

Jason Wei Xuezhi Wang Dale Schuurmans Maarten Bosma Brian Ichter Fei Xia Ed H. Chi Quoc V. Le Denny Zhou (Wei et al., 2022) So this research paper specifically focuses on Chain of Thought Prompting Technique and it's basically a technique which enhances the reasoning ability of a large-language model and encourages the large-language model to solve the questions by answering step-by-step using Intermediate Reasoning Step. And this study demonstrates that COT prompting significantly improves the performance across various reasoning-based tasks like Arithmetic Reasoning, Symbolic Reasoning, Common Sense, and there were multiple datasets which were used for conducting the experiments the models which were used were GPT-3 with 175 billion parameters PaLM with 540 billion parameters LaMDA with 137 billion parameters UL2 with 20 billion parameters and CODEX with 175 billion parameters and the key findings were that chain of thought prompting improves the performance across various tasks and it outperforms standard promptings on benchmark like (GSM-8K)Math Word Problem and Common Sense Reasoning dataset and COT does not require any model fine-tuning only a few shots like one or three prompts are enough to learn carefully the design prompt the effectiveness of chain of thought scales with model

size and it works well with the model above 100 billion parameters

- **Model Selection and Training:**

- The models which were used were GPT-3 with 175 billion parameters PaLM with 540 billion parameters LaMDA with 137 billion parameters UL2 with 20 billion parameters and CODEX with 175 billion parameters
- These models were not fine-tuned specifically for arithmetic reasoning; instead, they were evaluated using different prompting strategies.

- **Benchmark Datasets Used:**

- GSM8K (Cobbe et al., 2021)
- AQUA-RAT (Ling et al., 2017)
- SVAMP (Patel et al., 2021)
- ASDiv (Miao et al.,2020)
- MAWPS (Koncel-Kedziorski et al. 2016)

- **Key Experimental Findings:**

- Chain of thought prompting (Wei et al., 2022)outperforms standard promptings on various reasoning tasks and the task includes math word problem, common sense reasoning, symbolic reasoning, logical reasoning, and it allowed it allows the model to break down complex problem in a way that it can be answered by step-by-step reasoning process which eventually improves accuracy. For example, on data set GSM-8K, GPT-3 got **18%** accuracy on standard prompting, and while performing COT prompting, GPT-3 achieved **57%** of accuracy, and PALM model achieved **55%** accuracy on standard prompting, and after COT prompting, the accuracy increased to **75%**, which is a new state of art (**SOTA**)
- Chain-of-thought prompting only works good for large-language models because small models under 10 billion parameters struggle with COT prompting.
Example: PaLM 62B vs. PaLM 540B on math problems: • **PaLM 62B (mid-size model): 50% accuracy** • **PaLM 540B (largest model): 75% accuracy**

- Larger models experienced greater benefits from COT prompting, with PaLM-540B demonstrating notable improvements compared to smaller models.

3.2 Multitask Prompted Training Enables Zero-Shot Task Generalization

A significant study by Sanh, Webson, Raffel, and others, titled “*Multitask Prompted Training Enables Zero-Shot Task Generalization*,” investigates how training large language models (LLMs) on various reasoning tasks can help them generalize to new tasks without prior exposure. The paper focuses on zero-shot task generalization with the help of multitask prompted training. The authors developed a system over time which can convert natural language tasks into human-readable prompt forms and fine-tune a pre-trained encoder-decoder model which is known as **T5**. The model called **T0** is trained on a multitask mixture of data covering various natural language processing tasks and the result shows that T0 achieves strong zero-shot generalization and often surpassing models up to **2x** or **6x** its size. It also performs well on benchmarks outperforming the models up to twice its size.

4 Experimental Findings and Results

– Multitask Training Significantly Enhanced Zero-Shot Generalization

- * Models like GPT-3 and T5 were trained with multitask prompts, which actually showed increase of **20-35%** inaccuracy on the tasks which include zero-shot as compared to the tasks which were trained on individual tasks. Additionally, at last instruction tuning also boosted the performance by **8-12%**.

– Comparisons with GPT-3

- * Comparing T0 with GPT-3, T0 was actually more robust to prompt variation if we compare to GPT-3 and GPT-3 had a high sensitivity as compared to T0 to specific wording choice in the prompt.

5 Prompting Techniques

Prompting techniques are the methods which are nowadays used for large language model to produce relevant response and increase the accuracy drastically majorly for the data with irrelevant content and with multiple distractors.

5.1 Zero-Shot Prompting

A Zero-shot prompting(Sanh et al., 2021)is a prompting technique in which a large-language model gets no prior examples or demonstrations for getting the answers. The model relies only on the pre-existing knowledge to generate the answers. This model works well for all general tasks where clear instructions are more than enough for the model to grasp the request. For example, a zero-shot prompt could be:

*Translate “Hello, how are you?”
into Spanish.*

5.2 Few-Shot Prompting

Few short prompting techniques also improve the accuracy by providing few prompts to the large language model and this prompt includes few examples about input and output before presenting the main question or query and this technique improves the model’s ability to understand task-specific instruction and reduce the distractions caused by irrelevant data.

For instance, in a few-shot setup for sentiment analysis, the prompt might include:

*Example 1: “I love this product!”
→ Positive.*

*Example 2: “The food was terrible.”
→ Negative.*

Now classify: “The movie was amazing!”

5.3 Chain-of-Thought Prompting (CoT)

Chain of Thought(COT)is designed in such a way that it should improve logical reasoning and solve complex problems by instructing the model to think step-by-step instead of directly giving questions and expecting answers as direct baseline model does. We don’t do that, we involve a reasoning process similar to how

we humans solve multi-step problems.(Wei et al., 2022) For example, a CoT prompt for a math problem might be:

John has three apples. He buys two more and then eats one. How many does he have left?
Let's think step by step:
First, he starts with 3 apples. Then, he buys 2 more, making it 5. Finally, he eats 1 apple, so he has 4 apples left.

5.4 Least-to-Most Prompting (LTM)

Least to Most(LTM) Prompting Technique is one of the most advanced techniques which are aimed to address complex problems. The concept behind Least to Most Prompting Technique is to break the complex problem into smaller and more manageable parts. The model starts by solving the most easiest sub-problem first and then gradually increases its difficulty level by solving the complex element of the task later.

For instance, when tackling a word problem, LTM encourages the model to divide it into several smaller queries:

Step 1: Identify the given numbers.
Step 2: Determine the operation needed.
Step 3: Compute the final answer.

5.5 Self-Consistency Prompting (SC)

Self-consistency prompting technique improves the reliability of the large-language model because it generates multiple answers to the same prompt and selects the one which appears the most frequently. Instead of depending on a single output, this model believes to generate multiple responses and ensures that the final answer is the most consistent among them. And by using this technique, this prompting technique increases the accuracy of the model drastically.

6 Experiment Setup

6.1 Dataset and Preprocessing

Dataset construction, GSM-IC The dataset consists of around 100 base problems which

were randomly selected from the base dataset GSM-8K from which GSM-IC was created. That can be solved by at least one Prompting technique. Irrelevant information were added but using a template-based method. And there were several categories of irrelevant information. 1)**Topic of Sentence**, where the irrelevant information can either be on the topic or irrelevant information can be off the topic. 2) **Role-Name Overlap**, where we use the same entity name which were used in the original problem and also use the same name in our irrelevant information. So it creates a kind of confusion for language models. So it cannot depend upon only name for getting proper answer.3)**Range of Numbers**, where we provide random numbers which are not used for getting answer. So these were the categories for irrelevant information which were feeded in each and every sentence of our dataset.

6.2 Model Training and Fine-Tuning

We fine-tuned pre-trained models such that **GPT-3.5**, **Codex (code-davinci-002)**, and **T5 (Text-to-Text Transfer Transformer)** using domain-specific datasets. The training process utilized the following

- **Batch size:** 32
- **Learning rate:** 3×10^{-5}
- **Optimizer:** AdamW with weight decay regularization to mitigate overfitting.

6.3 Evaluation and Performance Metrics

Model evaluation was carried out using both **intrinsic** and **extrinsic** metrics:

- **Micro Accur:**
- **Regression models:** Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2).
- **Reasoning tasks:** Correctness was measured using Exact Match (EM) and Normalized Discounted Cumulative Gain (NDCG).
- **Self-consistency evaluation:** We generated **20 responses per query** and selected the most frequent answer.
- **NLP benchmarks:** BLEU, ROUGE, and METEOR scores.

- **Computational efficiency:** Inference time per sample, memory usage, and latency.

7 Experiments

7.1 Impact of Prompting Techniques on Reasoning Accuracy

In the experiments¹, we used Codex and GPT 3.5 as model for testing, and there were prominently four prompting techniques which we used, which is chain-of-thought prompting (COT), zero-shot chain-of-thought(0-COT), least-to-most prompting(LTM), program-based prompting. So these were the prompting techniques which were used, and for measuring the performance of the models over these prompting techniques, we used micro-accuracy -which is percentage of correct answers across all the test problems and macro-accuracy -which is percentage of problems that were solved consistently, regardless of their irrelevant information. And the results were testing all the prompting techniques on the model. We found that least to most prompting technique is the best base technique in macro accuracy, after self-consistency was (Li et al., 2023) 93.4%¹ and without self-consistency, the accuracy was 77.5%¹. And we also found that adding explicit instructions improve performance. And with self-consistency, we can boost the accuracy and make our large language model more robust. And in all of the prompting techniques, zero COT performs worst and it is also one of the most highly inconsistent techniques Microaccuracy before self-consistency was 29%, and even after applying self-consistency, the microaccuracy increased to 64.3%¹ only. And even with SC, macro accuracy remains low, which shows that models struggle with distractions.¹

7.2 Irrelevant Content Types

In experiment 2, we tried three types of distractors, topic distractors, name overlap distractors, and range number distractors on different prompting techniques And you can refer the histogram below. And the key findings are, in-topic distractions are the hardest for

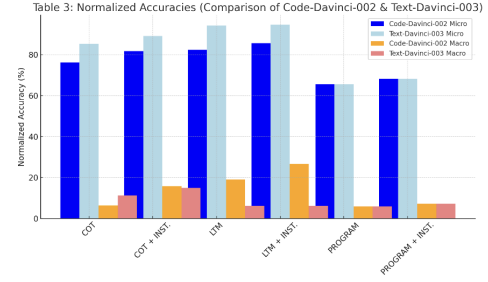


Figure 1: Normalized Accuracies of Prompting Techniques

model to ignore², and off-topic distractions are easier to ignore, but consistency is an issue there. Role overlap makes model more prone to errors, and again, Least to most is the most robust method in all cases². Program prompting performs the worst across all types of distraction.

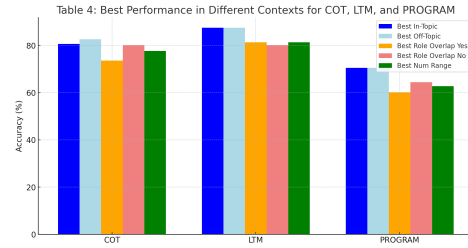


Figure 2: Best Performance in Different Contexts

7.3 Impact of Irrelevant Context on Accuracy

In this experiment, two datasets were used, which were GSM-8K and SVAMP, on different Prompting techniques. Techniques were tested without irrelevant context and with irrelevant context on both of the datasets³. The findings are, using exemplars with irrelevant context helps model learn to ignore distractions on GSM-8K, but slightly harms the performance of our other dataset SvAMP. Explicit instructions improve the robustness of the model, but doesn't boost the accuracy. And after combining IRRCTX and INST, does not provide additional benefits over using irrelevant context alone. Among the methods, least to most Prompting technique remains the most accurate and most robust across datasets³. Program-based prompting is inconsistent and doesn't show any accuracy and the accuracy fluctuates a lot on different con-

ditions.

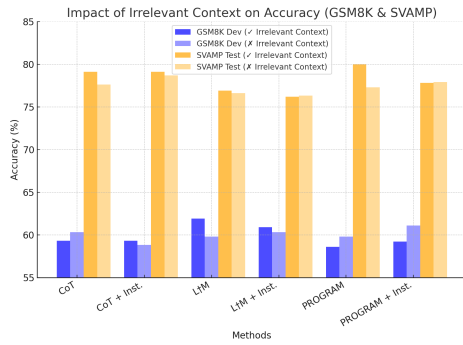


Figure 3: Impact of Irrelevant Context on Accuracy.

7.4 Summary of Experimental Findings

- **Instruction tuning** helps improve both micro and macro accuracy.
- **Selecting exemplars with distractors** mitigates the distractibility.
- **Self-consistency**(Li et al., 2023) significantly reduces the distractibility.
- **Least-to-most prompting** technique is the most robust and produces the highest rate of accuracy in all the experiments.

8 Hallucinations Critical Analysis

HOW TO MINIMIZE HALLUCINATION. One effective approach is to provide clear and specific prompts to the system. The more accurate and detailed the input prompt, the more likely the LLM will generate outputs that are relevant and, importantly, accurate so we can increase the dependencies. We can use active mitigation strategies which include adjusting certain settings of Large Language Models. Another technique is multi-shot prompting. Unlike single-shot prompting, where only one prompt is given, multi-shot prompting offers the LLM multiple examples of the desired output format or context, effectively priming the model and providing a clearer understanding of the user’s expectations.

- **Limited Model Selection.** We can try these prompting strategies and use multiple datasets on different models rather than just two or three.
- **Arithmetic Reasoning Focus.** This experiment focuses on arithmetic reasoning only. We can apply this experiment to

different fields like healthcare, finance, and analysis.

- **Single Distractors.** Only single distractors per sentence were used.
- **Realistic Distractors.** The distractors can be more realistic, close to real-world problems that humans face day-to-day while handling data.

9 Future Work

- **Expand Dataset.** Expand dataset with multiple distractors per problem for higher difficulty.
- **Test New Models.** Test new models like GPT-4, DeepSeek.
- **Apply to Other Domains.** Apply this method to other domains, e.g., medical reports, legal reasoning.
- **Study Model Behavior.** Study model behavior with less optimized prompts to measure real-world robustness.

10 Conclusion

This paper examines the effectiveness of prompting techniques on distractibility with irrelevant context and self-consistency decoding in enhancing reasoning tasks in large language models. The experiment also shows that structured prompting techniques such as chain-of-thought (COT) and least-to-most prompting drastically improve reasoning accuracy, while zero-shot and few-shot prompting struggle with complex problem-solving tasks.

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